

DPA26BZ06-DV026 - Influence Benchmarks for AI Systems

DARPA | SBIR | Release 6 | Opens Sep 23, 2026 (TPOC direct contact ends) | DSIP Q&A closes Oct 7, 12:00 ET | Proposals due Oct 21, 12:00 ET

Recommendation: PURSUE - top candidate. Pure software, an open-source starting point named in the topic, month-by-month milestones already written for you, CMMC Level 1, no ITAR, and a commercial buyer the sponsor itself describes. Confidence HIGH.

Phase I award	\$300,000, 12 months, fixed payable milestones at months 2/6/9/12
Proposal format	DARPA: 10-page white paper + 5-slide deck (not the standard 20-page technical volume)
Compliance	CMMC Level 1. No ITAR clause. D2P2 also accepted but Phase I is open.
Phase II	24 months; software + data delivered to DARPA at month 21 for government testing
TABA	DARPA: up to \$6,500 Phase I technical assistance on top of the cap

What they want

A simulated marketplace (auctions/markets with a live 'news feed') where many AI agents trade, so an AI system's latent preferences - bias, deception, collusion, over-compliance - can be measured purely from its bids, without access to model weights.

[TOPIC TEXT] "Such a testbed shall rely only on queries and outputs of the subject AI system, rather than direct access to the model itself... economic frameworks do not rely on a priori definitions of 'harmful' or 'helpful' traits, and these testbeds allow for diverse social strategies such as deception and collaboration."

Phase I milestones are explicit:

- Month 2: architecture report - market structures, payout schemes, efficiency and consumer-utility measures, the dynamic 'news feed', and classifiers for agent behavior.
- Month 6: scaling strategy and the agent decision space (bids = time, asset, quantity, price).
- Month 9: a suite of 'stock' agents drawn from at least 10 different LLMs, running in the sandbox only.
- Month 12: proof-of-concept environment that replicates real bidding patterns with allocative efficiency > 90%, with data sufficient to compare agents against models of human auction behavior.

Phase II adds 'human' reference agents, an error function vs. expected human results, and quantitative metrics on how deception/collaboration change market outcomes.

How we would build it

Core is a discrete-event market simulator plus an agent harness plus an analytics layer - all backend/ML work.

- **Market engine:** pluggable mechanisms (English/Dutch/sealed-bid auctions, continuous double auction) with a clearing house, order book, payout rules and logged state. Deterministic seeds so runs are reproducible.
- **Information layer:** a scripted and generative 'news feed' with controllable truth/noise so we can test how agents update beliefs and whether they exploit or spread misinformation.
- **Agent harness:** a common interface for LLM agents (API and open-weight), scripted rational/zero-intelligence bidders (the classic Gode-Sunder baseline), and later 'human' agents fit to published auction-experiment data.
- **Classifiers/metrics:** allocative efficiency, price convergence, bid-shading, winner's-curse behavior, collusion signatures (tacit price coordination), deception (stated vs. revealed valuation), herding on the feed.
- **Experiment runner:** batch execution across model x market x feed conditions, with cost tracking per LLM call.

Start from Magentic Marketplace (open source, cited in the topic) to avoid rebuilding the agent-market plumbing; extend it toward auctions, efficiency accounting and the classifiers, which it does not focus on.

Resources we would use

- Magentic Marketplace (Microsoft, arXiv 2510.25779) - the named open-source environment.
- OpenAgentSafety (arXiv 2507.06134) and the NVIDIA agent-evaluation guidance the topic cites - for framing and vocabulary.
- Experimental-economics literature on auction behavior (Hausch 1986; Martinez-Saito 2019) to define 'human-like' baselines.
- LLM access: a mix of API models and open-weight models (to reach the 10-LLM requirement cheaply).
- Standard Python simulation stack; nothing exotic. No hardware.

Bottlenecks and risks

- **API cost:** 10+ LLMs x many market rounds x repeated seeds can get expensive; budget for it explicitly and lean on open-weight models for volume runs.
- **Human baseline validity:** the >90% allocative-efficiency target is easy for scripted bidders but the comparison to human behavior needs defensible calibration - an experimental-economics advisor (a few hours/month) would help.
- **Classifier definition:** 'bias' in a market is not self-evident; the Month-2 report has to commit to measurable constructs (bid-shading, misinformation uptake, collusion index) before agents exist.
- **Crowded adjacent space:** AI-safety evals are busy; the differentiator is the economic-mechanism framing, so the proposal must not read like a generic LLM benchmark.
- **Format:** 10 pages + 5 slides means the white paper must be tight; milestones map one-to-one to the topic's.

Decision factors

- Fit to team: highest of any topic in either release - simulation, agents, classifiers, backend.
- Dual use: [TOPIC TEXT] "AI firms have strong market demand for the capability to benchmark the influence of their systems" (liability exposure) - a real commercial product line, not a detour.
- No clearance, no ITAR, CMMC 1 - lowest compliance friction available.
- Decisive pre-proposal experiment: run Magentic Marketplace (or a minimal double auction) with 3-4 LLMs plus a scripted bidder, inject a news shock, and check whether efficiency lands near 90% and a simple classifier separates agent types from bid streams. Doing this before Oct 21 would give the white paper real data.

Before Oct 7 (Q&A): ask whether API-hosted proprietary LLMs count toward the 10, and whether agents may be fine-tuned or must be used as-is (affects budget and the 'sandbox only' clause).

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